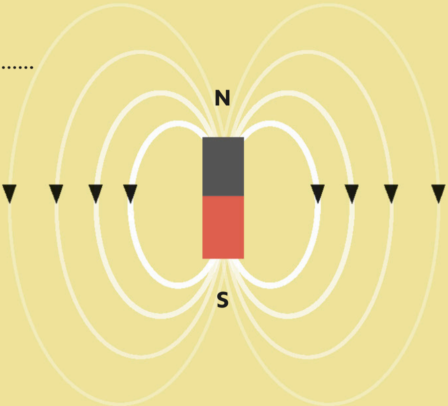
BAGNETIC DIPOLES

In this familiar bar magnet, countless dipoles (molecules with a positively and a negatively charged side) are aligned to produce a strong magnetic field.



Field lines

Magnetic fields are often depicted using lines that indicate the strength and direction of their influence.

FIELDS OF ATTRACTION

While electric charge is a fundamental property of many particles, magnetism is not—instead it is an effect that arises from electric charges in motion. Charged subatomic particles such as electrons and quarks develop their own weak magnetic fields, known as magnetic moments, as a consequence of their intrinsic angular momentum, or spin. Like spin and charge themselves, magnetic moments are quantized—only capable of taking on certain discrete values.

HAVING A MOMENT

An effect called electromagnetic induction means that any spinning electrically charged particle induces (creates) a magnetic field similar to that of a bar magnet.